## SOLOMON ISLANDS: MEDIUM-TERM SCENARIO AND DEBT SUSTAINABILITY ANALYSIS[1]

***If the government implements its reform agenda, the Solomon Islands could enjoy higher living standards over the medium term, although its debt situation remains fragile. If necessary reforms are not implemented, however, living standards and debt indicators could deteriorate further.***

### Medium-term scenario

**The staff's baseline medium-term framework is underpinned by policy adjustment.** It assumes that RAMSI and donors remain engaged, and although declining, aid flows continue to be substantial (9 percent of GDP in 2010 from 20 percent in 2005), and logging activity is halved. It also assumes that a critical mass of structural reforms is implemented so that new private activities take up the slack left by logging. Finally, it assumes that financial stability is maintained, underpinned by fiscal adjustment. In line with government policy, no new borrowing takes place.[2] However, downside risks to this scenario, particularly related to policy implementation, are high.

**Under these assumptions, real GDP growth is projected to average 4½ percent in 2006–10, driven largely by investment and exports** (Tables 5 and 6). Per capita income would rise by a cumulative 10 percent and inflation decline gradually to 5 percent as supply constraints ease. In spite of declining logging, exports grow strongly over the medium term, especially in 2006, as new export activities—primarily palm oil and mining—come on stream and expand. The current account deficit peaks (at around 13½ percent of GDP) in 2006 due to imported capital goods, and narrows over the projection horizon to around 4 percent of GDP. Reserves fall modestly as official development assistance (ODA) moderates and external debt repayments jump in 2008, but reserve coverage stabilizes at around 4 months of imports. The overall fiscal position is projected to improve resulting in modest budget surpluses over the medium term, with expenditure adjustment of 1½ percent of GDP and further revenue measures keeping the revenue-to-GDP ratio broadly constant, while import and logging duties fall by 2½ percent of GDP between 2005 and 2010.[3]

**If reforms stall causing donors and investors to leave, macroeconomic stability would likely falter** (Table 7). Under such an alternative scenario, growth is projected to slow to about 1½ percent, leaving real per capita income 6.5 percent lower by 2010 and almost 14 percent below the baseline. Inflation could gradually rise to 10 percent as supply

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[1] Prepared jointly with World Bank staff.

[2] Staff assumes no gross external borrowing throughout the projection horizon. Net domestic borrowing remains negative throughout the projection period.

[3] Staff assumes that the government gradually eliminates outstanding informal expenditure arrears by 2010, and realized contingent liabilities (of around 1.5 percent of GDP) to the NPF by 2015.

constraints reemerge. Production and exports weaken, and lower ODA and no FDI result in considerable import compression. With weaker growth limiting revenue, no revenue measures to compensate falling logging and import duties, and an inability to borrow, the government is forced to reduce recurrent spending by one fifth, with service provision falling further in line with lower ODA. Unpaid debt service adds to arrears, with government debt largely unchanged in 2010 from its level at end-2006.

**Debt sustainability in the baseline scenario**[4]

**The external debt sustainability analysis under the baseline scenario paints a mixed picture** (Annex I Table 1).[5] At end-2004, the NPV of external debt to GDP (at 45 percent) exceeded the Board-endorsed indicative threshold, while the ratios of the NPV of debt to exports and revenue were below their respective thresholds. However, with the recent realization of an external contingent liability worth 5 percent of GDP, the ratios of the NPV of debt to both GDP and exports are projected to exceed their respective thresholds at end-2005.[6]

**In the baseline adjustment scenario, all debt indicators improve over time.** The NPV of debt to exports is projected to fall below its threshold in 2006, but the NPV of debt to GDP still exceeds its threshold in 2010. Debt service to both exports and revenue (at 5.5 and 9.3 percent, respectively, in 2005) always remain below thresholds.

**The high level of domestic debt and informal expenditure arrears pose significant risks to fiscal sustainability** (Annex I Table 2). At end-2004, domestic debt was 30 percent of GDP (including expenditure arrears), bringing total public debt (including arrears) to 92 percent of GDP. Reflecting the realization of both the external contingent liability as well as domestic contingent liabilities worth 1.5 percent of GDP, public debt to GDP is projected to fall only slightly in 2005.[7] While all debt indicators decline under the baseline, the NPV of public debt remains above 100 percent of government domestic revenue and around 30 percent of GDP even in 2015.

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[4] Data on private external, state enterprise, and provincial government debt are not available. Therefore, the analysis is limited to central government debt data.

[5] This analysis assumes that the government remains current on all obligations from mid-2005, and that all debt in arrears is treated as a current short-term obligation with NPV equal to its face value. Existing external arrears are assumed to continue growing with accrued interest.

[6] The realized contingent liability relates to a government-guaranteed loan which the creditor called in during 2005. The loan was originally made to the government fishing company, Soltai, but was later transferred to the Investment Corporation of the Solomon Islands which has been unable to service it.

[7] The domestic contingent liabilities realized this year relate to loans made by the NPF to provincial governments and guaranteed by the central government.

**Risks to the debt dynamics**

**Historical scenarios and even some *temporary* stress tests highlight the fragility of the Solomon Islands' debt position** (Annex I Tables 3 and 4).

- *External Debt*: A return of key variables to their historical averages would see the ratio of the NPV of debt to exports and GDP rise during the projection period. Even a temporary fall in export growth would see the NPV of debt to exports jump and exceed the threshold almost throughout the period to 2025 and result in an external debt service ratio above its threshold for two years before 2010. A temporary fall in GDP growth would keep the NPV of debt to GDP above its threshold past 2010 and the NPV of debt to exports above its threshold to 2008. The concentration of exports, and their historical volatility, suggest these are very relevant stress tests for assessing vulnerability.[8]

- *Public Debt*: Lower GDP growth, even temporarily, and a return of growth and the primary balance to their historical averages result in the ratios of the NPV of public debt to GDP and revenue rising.

**These results are indicative only, as major uncertainties surround critical variables.** In particular, macroeconomic variables (especially GDP) are subject to large measurement errors. Data on aid flows are extremely limited, and staff estimates, based on discussions with donors, are subject to great uncertainty as future ODA plans are sketchy.

**Contingent liabilities—currently estimated at 10 percent of GDP—are another key source of uncertainty**. If all were realized as external debt, the NPV of debt would exceed 30 percent of GDP even in 2015. Additional contingent liabilities are likely to be uncovered as audits of state enterprise and provincial government accounts are underway for the first time in many years. Since state enterprises and provincial governments have few resources, a large share of any contingent liabilities related to them is likely to be realized.

**The Solomon Islands is at high risk of debt distress.** Some external debt indicators are currently above thresholds and domestic debt, arrears, and contingent liabilities are also high. Although debt indicators fall over the medium term under the staff's adjustment scenario, historical scenarios and several stress tests, even some temporary ones, lead to a deterioration in the debt burden. These tests seem especially relevant given the Solomon Islands' volatile economic history, and the nascent stage of the current reforms and recovery process. These findings, together with the country's poor credit history, suggest that the Solomon Islands is at high risk of debt distress.

[8] The coefficient of variation of export growth was almost -6 over the last ten years, with roughly the same volatility in absolute terms over the period since 1978.

Annex I. Table 1. Solomon Islands: External Debt Sustainability Framework, Baseline Scenario, 2002–25 1/
(In percent of GDP, unless otherwise indicated)

| | Actual | | | Historical Average 5/ | Standard Deviation 5/ | Estimate | Projections | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2002 | 2003 | 2004 | | | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2005-10 Average | 2015 | 2025 | Board Endorsed Threshold 6/ |
| **External debt (nominal)** 1/ | **67.0** | **70.6** | **62.0** | | | **61.4** | **57.1** | **53.0** | **48.1** | **44.2** | **40.4** | | **26.0** | **13.4** | |
| o/w public and publicly guaranteed (PPG) | 67.0 | 70.6 | 62.0 | | | 61.4 | 57.1 | 53.0 | 48.1 | 44.2 | 40.4 | | 26.0 | 13.4 | |
| Change in external debt | 18.0 | 3.6 | -8.6 | | | -0.6 | -4.2 | -4.1 | -4.9 | -4.0 | -3.7 | | -2.3 | -0.7 | |
| Identified net debt-creating flows | 16.6 | -6.1 | -25.5 | | | -1.4 | -2.1 | -2.3 | -3.0 | -4.7 | -4.2 | | -2.2 | -2.2 | |
| **Non-interest current account deficit** | **5.8** | **-2.4** | **-13.5** | **1.3** | **7.0** | **9.8** | **13.0** | **8.5** | **6.5** | **4.2** | **3.6** | | **2.5** | **-0.2** | |
| Deficit in balance of goods and services | 2.2 | 2.2 | -2.2 | | | 13.2 | 17.1 | 12.0 | 10.0 | 7.6 | 7.0 | | 5.6 | 2.6 | |
| Exports | 32.9 | 41.9 | 49.2 | | | 45.4 | 48.2 | 47.4 | 46.9 | 47.0 | 46.6 | | 42.1 | 38.9 | |
| Imports | 35.1 | 44.1 | 47.0 | | | 58.6 | 65.4 | 59.4 | 56.9 | 54.6 | 53.6 | | 47.6 | 41.5 | |
| Net current transfers (negative = inflow) | 2.4 | -5.9 | -9.8 | -2.9 | 4.7 | -4.0 | -3.9 | -3.7 | -3.6 | -3.6 | -3.5 | | -3.2 | -2.8 | |
| Other current account flows (negative = net inflow) | 1.3 | 1.2 | -1.5 | | | 0.5 | -0.3 | 0.3 | 0.2 | 0.2 | 0.1 | | 0.1 | 0.0 | |
| **Net FDI and other non-debt-creating flows (negative = inflow)** | **-1.6** | **-5.4** | **-4.9** | **-1.9** | **1.9** | **-10.3** | **-13.4** | **-9.5** | **-8.4** | **-7.9** | **-6.9** | | **-4.3** | **-2.0** | |
| **Endogenous debt dynamics** 2/ | **12.3** | **1.7** | **-7.0** | | | **-0.8** | **-1.6** | **-1.2** | **-1.2** | **-1.0** | **-0.9** | | **-0.4** | **0.1** | |
| Contribution from nominal interest rate | 1.9 | 2.2 | 1.3 | | | 1.7 | 1.2 | 1.1 | 1.0 | 1.0 | 0.9 | | 0.7 | 0.6 | |
| Contribution from real GDP growth | 1.4 | -3.7 | -3.4 | | | -2.5 | -2.7 | -2.3 | -2.2 | -2.0 | -1.8 | | -1.1 | -0.5 | |
| Contribution from price and exchange rate changes | 9.0 | 3.3 | -4.9 | | | … | … | … | … | … | … | | … | … | |
| **Residual (3-4)** 3/ | **1.5** | **9.7** | **16.8** | | | **0.8** | **-2.2** | **-1.9** | **-1.9** | **0.7** | **0.5** | | **-0.1** | **1.5** | |
| o/w exceptional financing | -1.4 | -1.3 | -1.4 | | | -1.0 | -0.5 | -0.5 | -0.5 | -0.5 | -0.5 | | -0.5 | -0.5 | |
| NPV of external debt | ... | ... | 45.0 | | | 46.5 | 43.5 | 40.8 | 37.6 | 34.9 | 32.4 | | 22.3 | 12.6 | |
| In percent of exports | ... | ... | 91.3 | | | 102.3 | 90.3 | 86.1 | 80.1 | 74.4 | 69.6 | | 52.9 | 32.4 | |
| **NPV of PPG external debt** | **...** | **...** | **45.0** | | | **46.5** | **43.5** | **40.8** | **37.6** | **34.9** | **32.4** | | **22.3** | **12.6** | **30** |
| **In percent of exports** | **...** | **...** | **91.3** | | | **102.3** | **90.3** | **86.1** | **80.1** | **74.4** | **69.6** | | **52.9** | 32.4 | **100** |
| **In percent of revenue (excluding grants)** | **...** | **...** | **168.1** | | | **174.5** | **161.3** | **151.3** | **138.7** | **129.3** | **120.0** | | **82.4** | ... | **200** |
| Debt service-to-exports ratio (in percent) | 4.9 | 6.2 | 4.4 | | | 5.5 | 4.8 | 4.6 | 5.1 | 4.7 | 4.5 | | 3.3 | 1.4 | |
| **PPG debt service-to-exports ratio (in percent)** | 4.9 | 6.2 | **4.4** | | | **5.5** | **4.8** | **4.6** | **5.1** | **4.7** | **4.5** | | **3.3** | 1.4 | **15** |
| **PPG debt service-to-revenue ratio (in percent)** | 10.0 | 13.2 | **8.1** | | | **9.3** | **8.6** | **8.2** | **8.9** | **8.2** | **7.8** | | **5.1** | 2.1 | **25** |
| Total gross financing need (billions of U.S. dollars) | 0.0 | 0.0 | 0.0 | | | 0.005 | 0.006 | 0.004 | 0.002 | -0.005 | -0.005 | | -0.002 | -0.015 | |
| Non-interest current account deficit that stabilizes debt ratio | -12.2 | -6.0 | -4.9 | | | 10.3 | 17.2 | 12.6 | 11.4 | 8.2 | 7.3 | | 4.8 | 0.5 | |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | -2.4 | 5.6 | 5.5 | -0.5 | 10.8 | 4.4 | 4.8 | 4.4 | 4.4 | 4.4 | 4.4 | 4.5 | 4.0 | 3.8 | |
| GDP deflator in US dollar terms (change in percent) | -15.5 | -4.6 | 7.4 | -1.0 | 9.1 | 5.8 | 2.5 | 2.3 | 2.5 | 1.5 | 1.6 | 2.7 | 1.9 | 1.4 | |
| Effective interest rate (percent) 4/ | 3.2 | 3.3 | 2.0 | 2.1 | 1.1 | 3.1 | 2.0 | 2.1 | 2.1 | 2.1 | 2.2 | 2.3 | 2.6 | 4.2 | |
| Growth of exports of G&NFS (US dollar terms, in percent) | 3.7 | 28.1 | 33.2 | -4.9 | 42.8 | 1.9 | 14.1 | 4.9 | 6.0 | 6.2 | 5.2 | 6.4 | 4.2 | 4.7 | |
| Growth of imports of G&NFS (US dollar terms, in percent) | -34.7 | 26.5 | 20.9 | -9.1 | 39.3 | 37.8 | 19.7 | -3.0 | 2.6 | 1.7 | 4.1 | 10.5 | 4.7 | 4.1 | |
| Grant element of new public sector borrowing (in percent) | ... | ... | ... | ... | ... | 50.1 | 50.1 | 47.9 | ... | ... | ... | 49.3 | ... | ... | |
| *Memorandum item:* | | | | | | | | | | | | | | | |
| Nominal GDP (billions of U.S. dollars) | 0.226 | 0.228 | 0.258 | | | 0.285 | 0.306 | 0.327 | 0.350 | 0.371 | 0.394 | | 0.526 | 0.882 | |

Sources: Data provided by the authorities; and Fund staff estimates.

1/ Refers to central government debt only due to data availability.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and $\rho$ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Current-year interest payments derived by previous period debt stock.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
6/ These thresholds apply to countries with weak policies and institutions and are discussed in *Operational Framework for Debt Sustainability Analysis in Low-Income Countries--Further Considerations*.

Annex I. Table 2. Solomon Islands: Public Sector Debt Sustainability Framework, Baseline Scenario, 2002–25

(In percent of GDP, unless otherwise indicated)

| | Actual | | | | Estimate | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2003 | 2004 | Historical Average 6/ | Standard Deviation 6/ | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2005-10 Average | 2015 | 2025 |
| **Public sector debt 1/** | 121.9 | 92.5 | | | 88.6 | 81.2 | 73.5 | 65.6 | 59.2 | 52.9 | | 32.5 | 15.4 |
| *Of which:* Foreign-currency denominated | 70.4 | 62.1 | | | 61.9 | 58.4 | 54.0 | 49.1 | 45.1 | 40.9 | | 26.3 | 13.6 |
| Change in public sector debt | 0.8 | -29.4 | | | -3.9 | -7.4 | -7.8 | -7.9 | -6.4 | -6.2 | | -3.5 | -0.9 |
| Identified debt-creating flows | -11.4 | -22.2 | | | -7.0 | -7.3 | -7.4 | -6.7 | -5.4 | -5.4 | | -3.5 | -0.2 |
| Primary deficit | -0.9 | -10.1 | 1.0 | 5.8 | -1.8 | -2.9 | -2.8 | -2.6 | -2.5 | -2.4 | -2.5 | -2.1 | -0.6 |
| Revenue and grants | 37.6 | 48.9 | | | 48.5 | 42.8 | 40.9 | 40.6 | 39.7 | 37.8 | | 34.4 | 31.3 |
| *Of which:* Grants | 18.1 | 22.1 | | | 21.8 | 15.8 | 14.0 | 13.5 | 12.7 | 10.8 | | 7.3 | 4.4 |
| Primary (noninterest) expenditure | 36.7 | 38.7 | | | 46.6 | 39.9 | 38.2 | 38.0 | 37.2 | 35.4 | | 32.3 | 30.7 |
| Automatic debt dynamics | -9.8 | -12.0 | | | -6.3 | -4.4 | -4.7 | -4.1 | -3.0 | -2.9 | | -1.3 | 0.3 |
| Contribution from interest rate/growth differential | -7.4 | -9.4 | | | -4.8 | -5.0 | -4.3 | -3.8 | -3.2 | -2.8 | | -1.3 | 0.2 |
| *Of which:* Contribution from average real interest rate | -1.0 | -3.0 | | | -0.9 | -0.9 | -0.9 | -0.7 | -0.5 | -0.3 | | 0.0 | 0.8 |
| *Of which:* Contribution from real GDP growth | -6.4 | -6.4 | | | -3.9 | -4.1 | -3.4 | -3.1 | -2.8 | -2.5 | | -1.4 | -0.6 |
| Contribution from real exchange rate depreciation | -2.4 | -2.6 | | | -1.5 | 0.5 | -0.3 | -0.3 | 0.2 | -0.2 | | ... | ... |
| Other identified debt-creating flows | -0.7 | 0.0 | | | 1.1 | 0.0 | 0.0 | 0.0 | 0.0 | -0.1 | | -0.1 | 0.0 |
| Privatization receipts (negative) | 0.0 | 0.0 | | | -0.4 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 |
| Recognition of implicit or contingent liabilities 2/ | 0.0 | 0.0 | | | 1.5 | 0.0 | 0.0 | 0.0 | 0.0 | -0.1 | | -0.1 | 0.0 |
| Debt relief (HIPC and other) | -0.7 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 |
| Other (specify, e.g., bank recapitalization) | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 |
| Residual, including asset changes | 12.2 | -7.3 | | | 3.1 | -0.1 | -0.3 | -1.2 | -1.0 | -0.9 | | 0.0 | -0.6 |
| **NPV of public sector debt** | 102.3 | 75.3 | | | 73.1 | 66.4 | 60.2 | 54.1 | 49.1 | 44.4 | | 28.5 | 14.4 |
| *Of which:* Foreign-currency denominated | 50.9 | 45.0 | | | 46.5 | 43.5 | 40.8 | 37.6 | 34.9 | 32.4 | | 22.3 | 12.6 |
| *Of which:* External | 50.9 | 45.0 | | | 46.5 | 43.5 | 40.8 | 37.6 | 34.9 | 32.4 | | 22.3 | 12.6 |
| NPV of contingent liabilities (not included in public sector debt) 2/ | ... | ... | | | 10.0 | 9.0 | 8.1 | 7.2 | 6.5 | 6.0 | | 3.9 | 1.8 |
| Gross financing need 3/ | 1.9 | -1.7 | | | 4.1 | 1.0 | 0.7 | 0.6 | 0.4 | 0.3 | | 0.2 | 0.3 |
| NPV of public sector debt-to-revenue ratio (in percent) 4/ | 523.6 | 281.5 | | | 274.6 | 245.7 | 223.3 | 199.7 | 181.6 | 164.4 | | 105.5 | 53.6 |
| *Of which:* External | 260.4 | 168.1 | | | 174.5 | 161.3 | 151.3 | 138.7 | 129.3 | 120.0 | | 82.4 | 46.9 |
| Debt service-to-revenue ratio (in percent) 4/ 5/ | 14.5 | 25.0 | | | 17.0 | 10.6 | 9.5 | 8.5 | 7.8 | 7.3 | | 4.8 | 2.3 |
| Primary deficit that stabilizes the debt-to-GDP ratio | -1.7 | 19.3 | | | 2.1 | 4.5 | 5.0 | 5.3 | 3.9 | 3.8 | | 1.4 | 0.3 |
| **Key macroeconomic and fiscal assumptions** | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 5.6 | 5.5 | -0.5 | 6.9 | 4.4 | 4.8 | 4.4 | 4.4 | 4.4 | 4.4 | 4.5 | 4.0 | 3.8 |
| Average nominal interest rate on forex debt (in percent) | 3.3 | 2.0 | 3.1 | 1.0 | 3.1 | 2.1 | 2.1 | 2.1 | 2.2 | 2.2 | 2.3 | 2.6 | 0.0 |
| Average real interest rate on domestic currency debt (in percent) | -4.3 | -5.4 | -3.6 | 2.8 | -4.1 | -3.7 | -4.5 | -4.3 | -3.3 | -2.6 | -3.7 | -1.8 | 29.3 |
| Real exchange rate depreciation (in percent, + indicates depreciation) | -3.4 | -4.0 | 3.9 | 12.2 | -2.5 | ... | ... | ... | ... | ... | ... | ... | ... |
| Inflation rate (GDP deflator, in percent) | 6.1 | 7.1 | 7.1 | 2.7 | 6.6 | 6.2 | 7.1 | 6.9 | 5.9 | 5.2 | 6.3 | 4.7 | 4.1 |
| Growth of real primary spending (deflated by GDP deflator, in percent) | 46.4 | 11.3 | 3.9 | 24.3 | 25.8 | -10.3 | -0.2 | 3.9 | 2.3 | -0.8 | 3.5 | 3.0 | 3.3 |
| Grant element of new external borrowing (in percent) | ... | ... | ... | ... | 50.1 | 50.1 | 47.9 | ... | ... | ... | 49.3 | ... | ... |

Sources: Data provided by the authorities; and Fund staff estimates.

1/ Gross central government debt, including informal arrears. Foreign currency denominated debt converted using end-of-period exchange rate.
2/ Current estimate of contingent liabilities recognition. Remaining contingent liabilities are estimated at 10 percent of GDP.
3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
4/ Revenues excluding grants.
5/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Annex I. Table 3. Solomon Islands: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005–25
(In percent)

| | Estimate | Projections | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2015 | 2025 |
| | | | | (NPV of debt-to-GDP ratio) | | | | |
| Baseline | 46 | 44 | 41 | 38 | 35 | 32 | 22 | 13 |
| A. Alternative Scenarios | | | | | | | | |
| A1. Key variables at their historical averages in 2006-25 1/ | 46 | 47 | 48 | 49 | 50 | 52 | 60 | 81 |
| A2. New public sector loans on less favorable terms in 2006-25 2/ | 46 | 44 | 41 | ... | ... | ... | ... | ... |
| A3. Realization of half of contingent liabilities as external debt in 2006 3/ | 46 | 48 | 46 | 42 | 40 | 37 | 26 | 16 |
| B. Bound Tests | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2006-07 4/ | 46 | 49 | 51 | 47 | 43 | 40 | 26 | 14 |
| B2. Export value growth at historical average minus one standard deviation in 2006-07 5/ | 46 | 54 | 66 | 61 | 58 | 55 | 42 | 18 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2006–07 | 46 | 49 | 52 | 48 | 44 | 41 | 26 | 14 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2006–07 6/ | 46 | 53 | 58 | 54 | 51 | 48 | 35 | 15 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 46 | 62 | 86 | 81 | 76 | 72 | 54 | 24 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 7/ | 46 | 61 | 57 | 52 | 48 | 44 | 28 | 13 |
| | | | | (NPV of debt-to-exports ratio) | | | | |
| Baseline | 102 | 90 | 86 | 80 | 74 | 70 | 53 | 32 |
| A. Alternative Scenarios | | | | | | | | |
| A1. Key variables at their historical averages in 2006–25 1/ | 102 | 98 | 101 | 104 | 107 | 112 | 144 | 208 |
| A2. New public sector loans on less favorable terms in 2006–25 2/ | 102 | 91 | 87 | ... | ... | ... | ... | ... |
| A3. Realization of half of contingent liabilities as external debt in 2006 3/ | 102 | 100 | 96 | 90 | 84 | 79 | 63 | 42 |
| B. Bound Tests | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2006–07 4/ | 102 | 102 | 109 | 100 | 92 | 86 | 61 | 36 |
| B2. Export value growth at historical average minus one standard deviation in 2006–07 5/ | 102 | 191 | 375 | 354 | 335 | 319 | 269 | 129 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2006–07 | 102 | 102 | 111 | 102 | 94 | 87 | 62 | 36 |
| B4. Net nondebt creating flows at historical average minus one standard deviation in 2006–07 6/ | 102 | 110 | 122 | 115 | 108 | 102 | 84 | 40 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 102 | 153 | 239 | 226 | 213 | 202 | 168 | 80 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 7/ | 102 | 127 | 120 | 111 | 102 | 95 | 67 | 33 |
| | | | | (Debt-service ratio) | | | | |
| Baseline | 5.5 | 4.8 | 4.6 | 5.1 | 4.7 | 4.5 | 3.3 | 1.4 |
| A. Alternative Scenarios | | | | | | | | |
| A1. Key variables at their historical averages in 2006–25 1/ | 5.7 | 5.4 | 5.6 | 6.7 | 6.6 | 6.9 | 7.8 | 10.8 |
| A2. New public sector loans on less favorable terms in 2006–25 2/ | 5.7 | 4.9 | 4.8 | 5.3 | ... | ... | ... | ... |
| A3. Realization of half of contingent liabilities as external debt in 2006 3/ | 5.7 | 5.4 | 5.2 | 5.6 | 5.1 | 4.9 | 3.7 | 1.7 |
| B. Bound Tests | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2006–07 4/ | 5.7 | 5.6 | 6.1 | 6.6 | 6.0 | 5.8 | 4.2 | 1.5 |
| B2. Export value growth at historical average minus one standard deviation in 2006–07 5/ | 5.7 | 8.4 | 13.9 | 16.6 | 15.3 | 14.6 | 11.0 | 10.5 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2006–07 | 5.7 | 4.9 | 4.8 | 5.2 | 4.7 | 4.5 | 3.3 | 1.4 |
| B4. Net nondebt creating flows at historical average minus one standard deviation in 2006–07 6/ | 5.7 | 4.9 | 5.1 | 5.8 | 5.4 | 5.1 | 3.8 | 3.0 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 5.7 | 6.9 | 9.4 | 11.0 | 10.1 | 9.7 | 7.3 | 6.4 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 7/ | 5.7 | 4.9 | 4.8 | 5.2 | 4.7 | 4.5 | 3.3 | 1.4 |
| Memorandum item: | | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) 8/ | 49 | 49 | 49 | 49 | 49 | 49 | 49 | 49 |

Sources: Data provided by the authorities; and Fund staff estimates.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and nondebt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.
3/ Assumes an interest rate equal to U.S. 6 month Libor + 2 percent.
3/ Liabilities realized and add to arrears from the beginning of 2006, attracting an interest rate equal to 2004 six month libor.
4/ Ratios calculated using baseline export-to-GDP, since GDP and exports are driven by the same primary commodities.
5/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
6/ Includes official and private transfers and FDI.
7/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
8/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

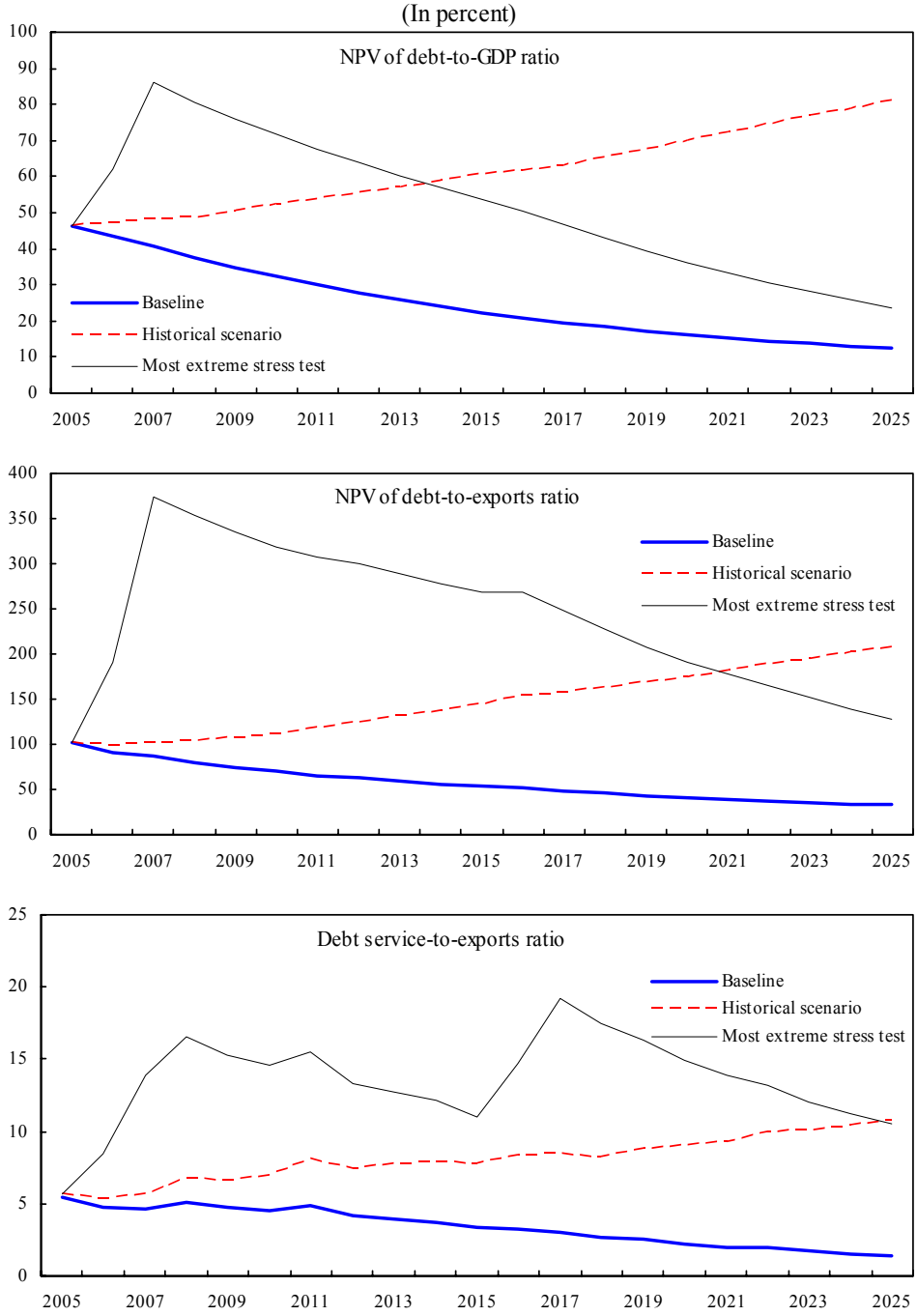
Annex I. Table 4. Solomon Islands: Sensitivity Analysis for Key Indicators of Public Debt, 2005–25

| | Estimate | Projections | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2015 | 2025 |
| | (NPV of Debt-to-GDP Ratio) | | | | | | | |
| Baseline | 73 | 66 | 60 | 54 | 49 | 44 | 29 | 14 |
| A. Alternative scenarios | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 73 | 74 | 74 | 74 | 75 | 76 | 88 | 116 |
| A2. Primary balance is unchanged from historical average | 73 | 70 | 67 | 64 | 62 | 60 | 57 | 54 |
| A3. Permanently lower GDP growth 1/ | 73 | 68 | 63 | 59 | 56 | 53 | 54 | 102 |
| B. Bound tests | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviation in 2006-2007 | 73 | 78 | 87 | 86 | 86 | 86 | 97 | 132 |
| B2. Primary balance is at historical average minus one standard deviation in 2006-2007 | 73 | 75 | 78 | 71 | 66 | 61 | 44 | 24 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 73 | 79 | 85 | 75 | 67 | 60 | 36 | 6 |
| B4. One-time 30 percent real depreciation in 2006 | 73 | 87 | 81 | 74 | 69 | 64 | 46 | 21 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2006 | 73 | 76 | 69 | 63 | 58 | 53 | 36 | 17 |
| | (NPV of Debt-to-Revenue Ratio) 2/ | | | | | | | |
| Baseline | 275 | 246 | 223 | 200 | 182 | 164 | 106 | 54 |
| A. Alternative scenarios | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 275 | 272 | 274 | 273 | 277 | 281 | 326 | 431 |
| A2. Primary balance is unchanged from historical average | 275 | 259 | 249 | 236 | 229 | 222 | 212 | 202 |
| A3. Permanently lower GDP growth 1/ | 275 | 251 | 234 | 217 | 207 | 198 | 201 | 378 |
| B. Bound tests | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviation in 2006-2007 | 275 | 290 | 323 | 317 | 318 | 319 | 358 | 493 |
| B2. Primary balance is at historical average minus one standard deviation in 2006-2007 | 275 | 279 | 288 | 262 | 243 | 225 | 162 | 88 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 275 | 292 | 314 | 278 | 249 | 223 | 134 | 24 |
| B4. One-time 30 percent real depreciation in 2006 | 275 | 324 | 299 | 273 | 254 | 235 | 168 | 79 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2006 | 275 | 280 | 256 | 232 | 213 | 195 | 135 | 63 |
| | (Debt Service-to-Revenue Ratio) 2/ | | | | | | | |
| Baseline | 17.0 | 10.6 | 9.5 | 8.5 | 7.8 | 7.3 | 4.8 | 2.3 |
| A. Alternative scenarios | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 17.0 | 11.2 | 11.3 | 11.4 | 11.6 | 12.0 | 13.9 | 34.1 |
| A2. Primary balance is unchanged from historical average | 17.0 | 10.6 | 10.2 | 9.9 | 9.6 | 9.5 | 9.0 | 15.5 |
| A3. Permanently lower GDP growth 1/ | 17.0 | 10.8 | 9.8 | 9.2 | 8.7 | 8.5 | 8.4 | 21.2 |
| B. Bound tests | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviation in 2006-2007 | 17.0 | 12.0 | 12.8 | 12.9 | 13.1 | 13.3 | 14.8 | 34.4 |
| B2. Primary balance is at historical average minus one standard deviation in 2006-2007 | 17.0 | 10.6 | 11.4 | 11.9 | 10.6 | 9.9 | 7.2 | 8.0 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 17.0 | 11.6 | 12.7 | 12.7 | 11.1 | 10.0 | 6.1 | 2.6 |
| B4. One-time 30 percent real depreciation in 2006 | 17.0 | 11.6 | 11.4 | 10.4 | 9.7 | 9.1 | 6.5 | 3.5 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2006 | 17.0 | 10.6 | 11.4 | 10.1 | 9.2 | 8.6 | 6.0 | 5.2 |

Sources: Data provided by the authorities; and Fund staff estimates.

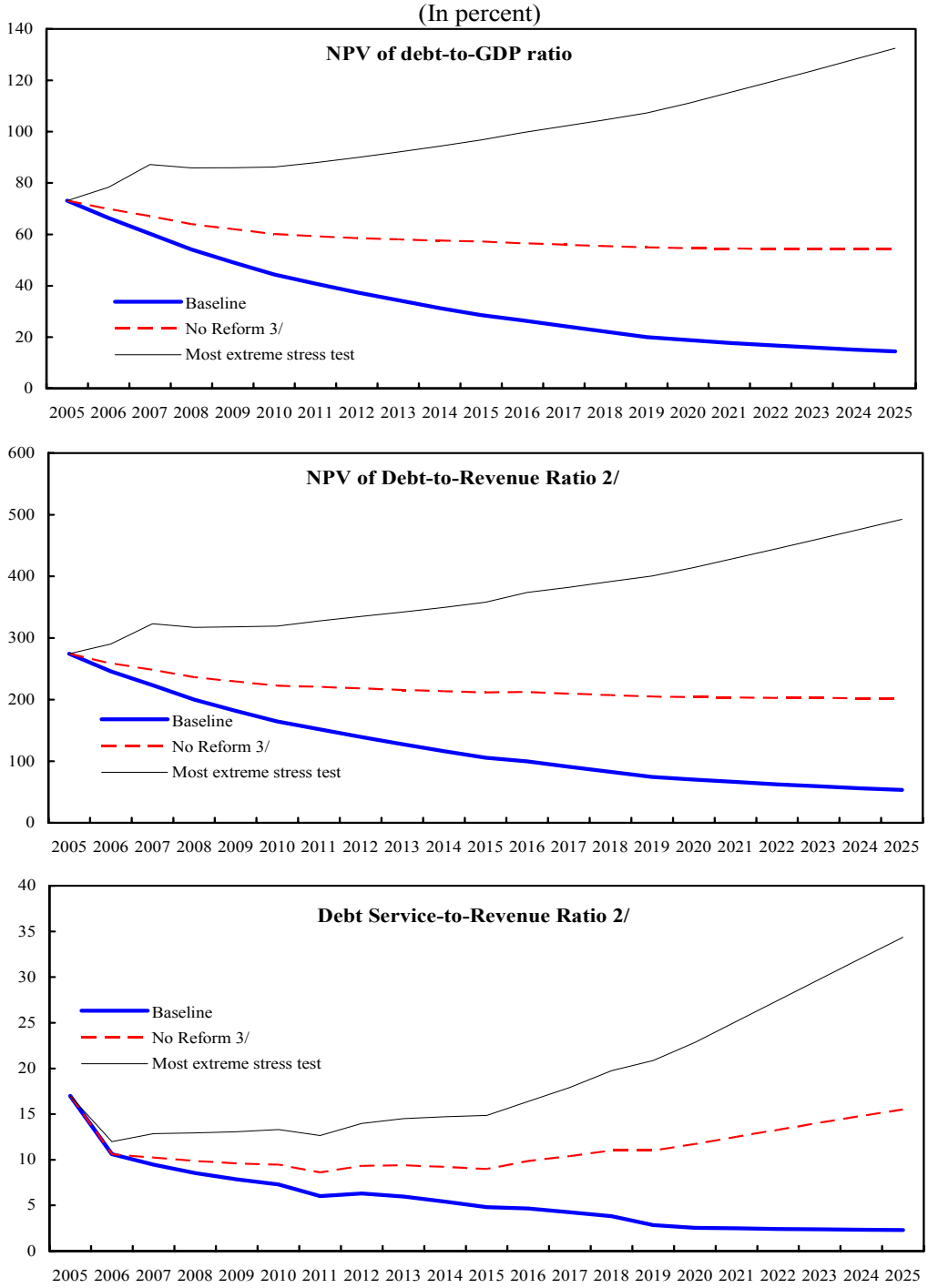
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined excluding grants.

Annex I Figure 1. Solomon Islands: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2005-2025
(In percent)




Source: Staff projections and simulations.

Annex I Figure 2. Solomon Islands: Indicators of Public Debt
Under Alternative Scenarios, 2005–25
(In percent)




Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2015.
2/ Revenue including grants.
3/ Primary balance remains at historical average.